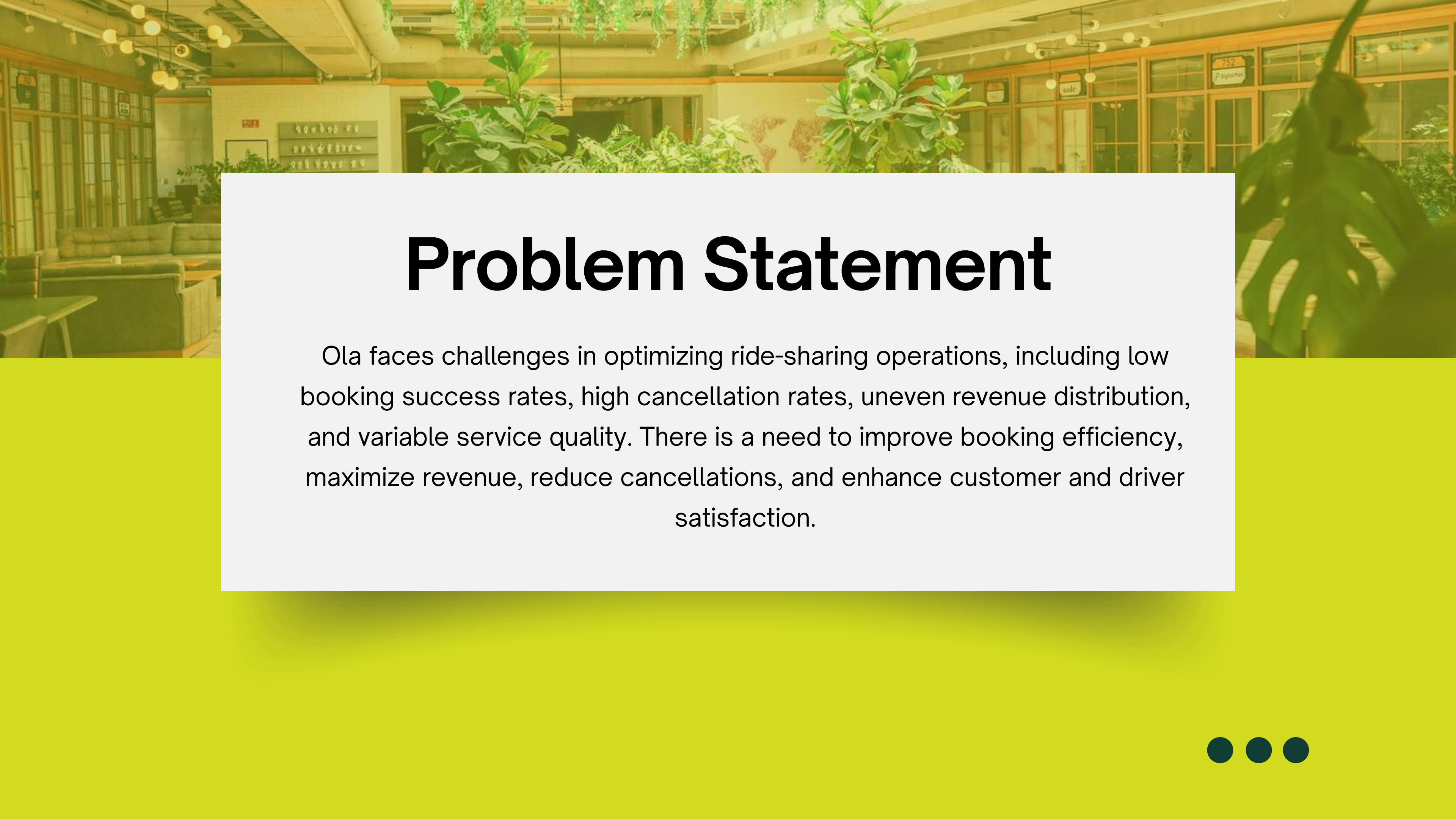




Ride Booking Analysis





Problem Statement

Ola faces challenges in optimizing ride-sharing operations, including low booking success rates, high cancellation rates, uneven revenue distribution, and variable service quality. There is a need to improve booking efficiency, maximize revenue, reduce cancellations, and enhance customer and driver satisfaction.



Goals

The goal is to improve Ola's booking efficiency, maximize revenue, reduce cancellations, and enhance customer and driver satisfaction through data-driven insights.

The analysis is structured into 5 key segments:



AD-HOC SQL QUERIES

- Successful bookings percentage
- Reasons for incomplete rides
- Customers with highest cancellations
- Average revenue by vehicle type



Query 1

Successful Booking %

Output

```
Successful Bookings Percentage

-- Step 1: Create a view for successful bookings
CREATE VIEW Successful_Booking AS
SELECT *
FROM Ola.bookings_data
WHERE Booking_Status = 'Success';

-- Step 2: Calculate the success percentage
SELECT
  ROUND(
    ((SELECT COUNT(*) FROM Successful_Booking) /
     (SELECT COUNT(*) FROM Ola.bookings_data) * 100), 2
  ) AS Success_percentage;
```

	Success_percentage
▶	62.09

✓ 62.09% of total bookings are successful.

Query 2

What is the main reason for incomplete rides?

Output

```
Incomplete Ride Reason

SELECT COUNT(*), Incomplete_Rides_Reason
FROM bookings_data
WHERE Incomplete_Rides = 'Yes'
GROUP BY Incomplete_Rides_Reason
ORDER BY COUNT(*) DESC;
```

COUNT(*)	Incomplete_Rides_Reason
1601	Customer Demand
1591	Vehicle Breakdown
734	Other Issue

- ✓ Customer demand is the top reason for incomplete rides, followed by issues like vehicle breakdown and other reasons.
- ✓ Recommendation: Make it easier for customers to reschedule instead of canceling, and ensure regular vehicle checks to avoid breakdowns.

Query 3

Which customer has the highest number of cancellations?

Output

Customer_ID	Cancellation_Count
CID460299	4

```
Customer with the Highest Cancellations|

SELECT Customer_ID, COUNT(*) AS Cancellation_Count
FROM bookings_data
WHERE Booking_Status IN ('Canceled by Customer', 'Canceled by Driver')
GROUP BY Customer_ID
ORDER BY Cancellation_Count DESC
LIMIT 1;
```

- ✓ Customer CID785112 has the highest cancellations, indicating frequent booking failures.
- ✓ Recommendation: Monitor such customers and apply stricter cancellation rules or send reminders to reduce booking failures.

Query 4

What Is the Average Revenue Generated by Each Vehicle Type?

```
Customer with the Highest Cancellations|  
  
SELECT Customer_ID, COUNT(*) AS Cancellation_Count  
FROM bookings_data  
WHERE Booking_Status IN ('Canceled by Customer', 'Canceled by Driver')  
GROUP BY Customer_ID  
ORDER BY Cancellation_Count DESC  
LIMIT 1;
```

Output

Vehicle_Type	Avg_Revenue_Per_Ride
Prime Sedan	556.99
Prime Plus	552.64
Auto	551.09
eBike	550.62
Bike	544.34
Prime SUV	542.11
Mini	540.72

- ✓ Most vehicle types generate similar average revenue per ride. Prime Sedans and Prime Plus have the highest, while Minis generate the least.
- ✓ Recommendation: Promote Prime Sedans and Prime Plus for higher revenue. Boost Mini vehicle bookings with targeted offers or discounts.

OVERALL ANALYSIS

Analyzed booking success rates, cancellation breakdown, weekday vs weekend ride trends, and identified top pickup and drop locations.



Booking Status Breakdown

- 62% success
- 17.89% cancelled by driver
- 10.19% cancelled by customer
- 9.83% driver not found

Insight:

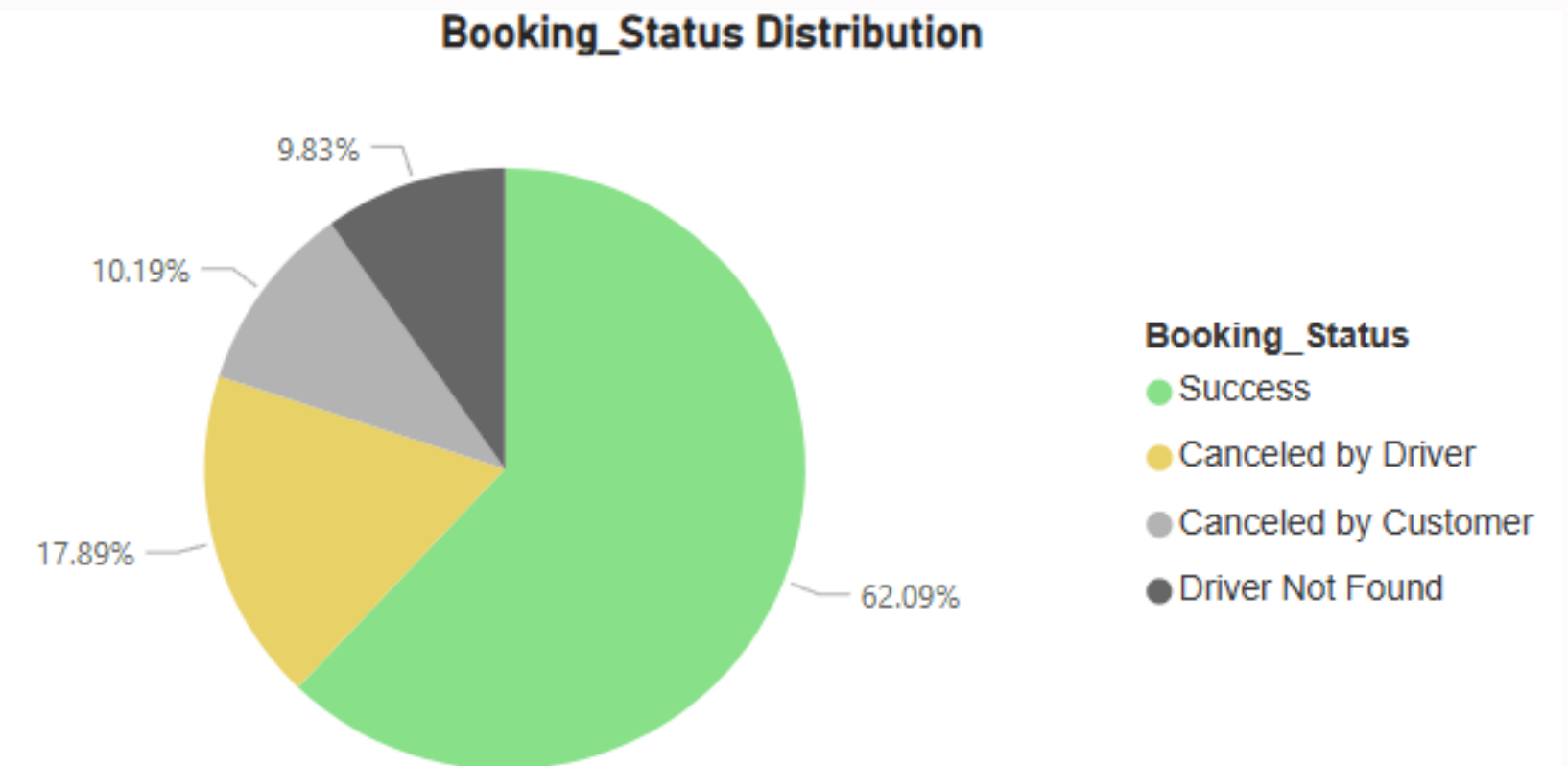
Most booking failures (about 28%) are because of drivers — either they cancel or no driver is found. Customer cancellations are lower.

Recommendation:

Motivate drivers with better incentives to accept rides.

Apply penalties if they cancel too often.

Also, improve how we assign drivers quickly, especially in busy hours.



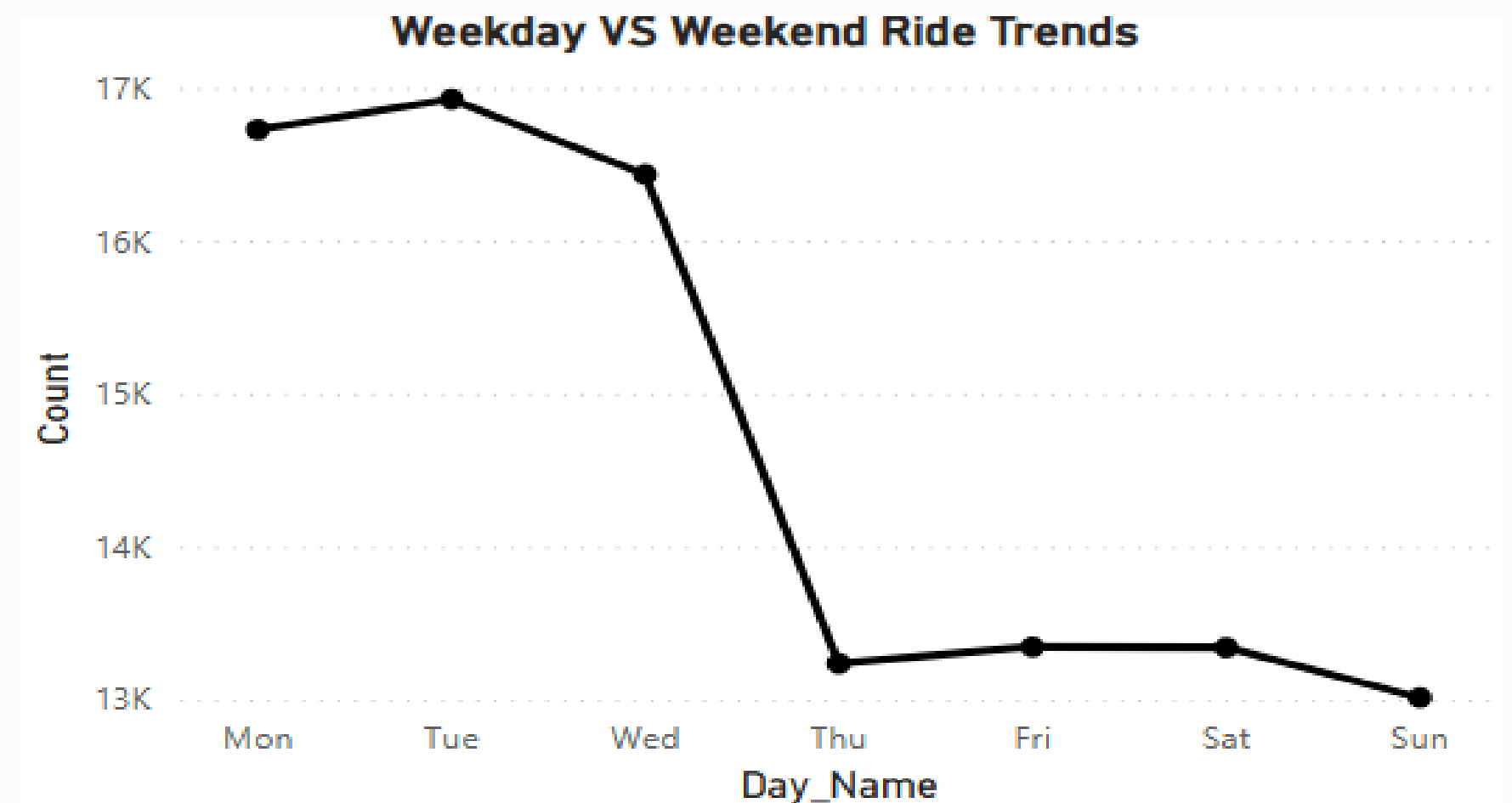
Weekday vs Weekend Ride Trends

Insight:

Ride demand is higher on weekdays compared to weekends. This shows the platform is mostly used for daily commuting on working days.

Recommendation:

Offer weekend promotions to boost ride bookings on those days.



Top Pickup & Drop Locations

Top Pickup Locations (Hot Zones)

Insight:

Banashankari, Yeshwanthpur , RT Nagar
These areas have the highest ride requests.

Recommendation:

Recommendation:
Ensure more drivers are active in these zones to reduce wait times and cancellations.

Top Drop Locations

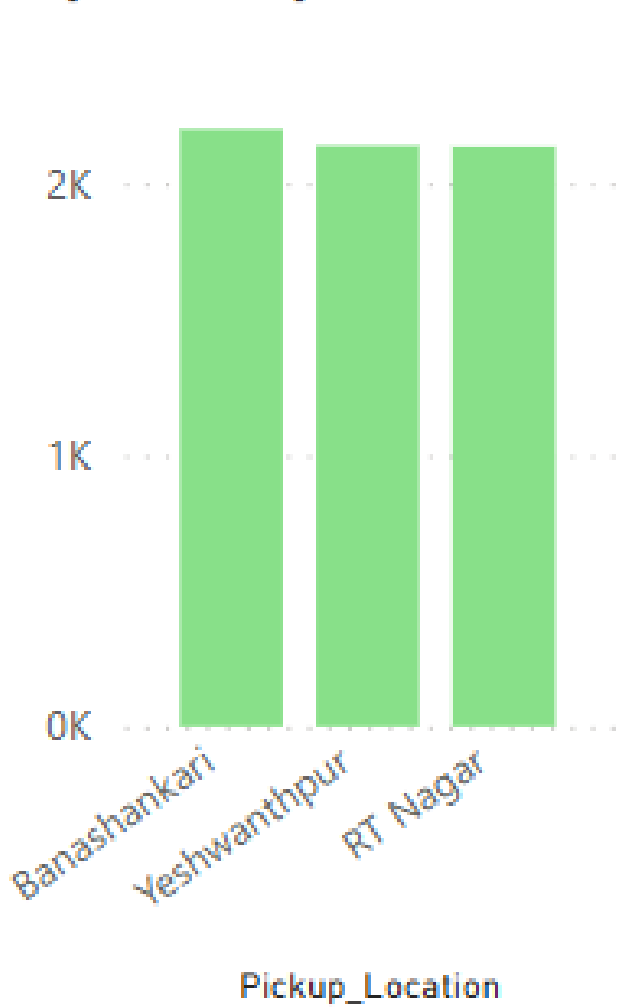
Insight:

Peenya , Mysore Road, MG Road
These are popular drop points, likely due to industries, offices, or shopping areas.

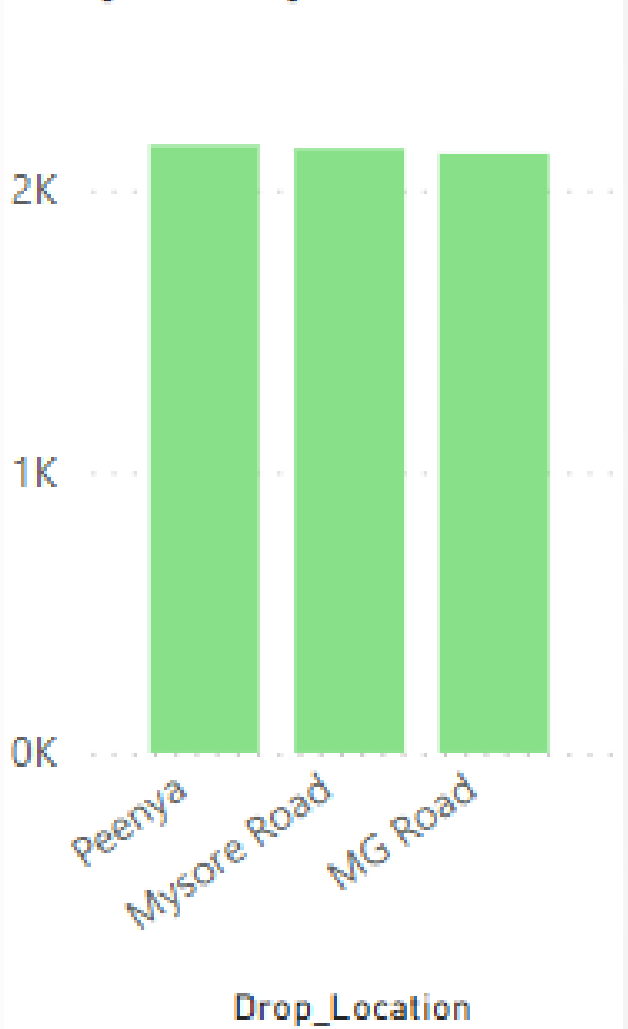
Recommendation:

Check for traffic or route issues in these areas.
Offer fixed fare deals on these routes to attract more bookings.

Top 3 Pickup Locations



Top 3 Drop Locations



VEHICLE TYPE ANALYSIS

Analyzed total bookings, revenue, ride distance, most booked vehicle types, and performance comparison across vehicle categories.



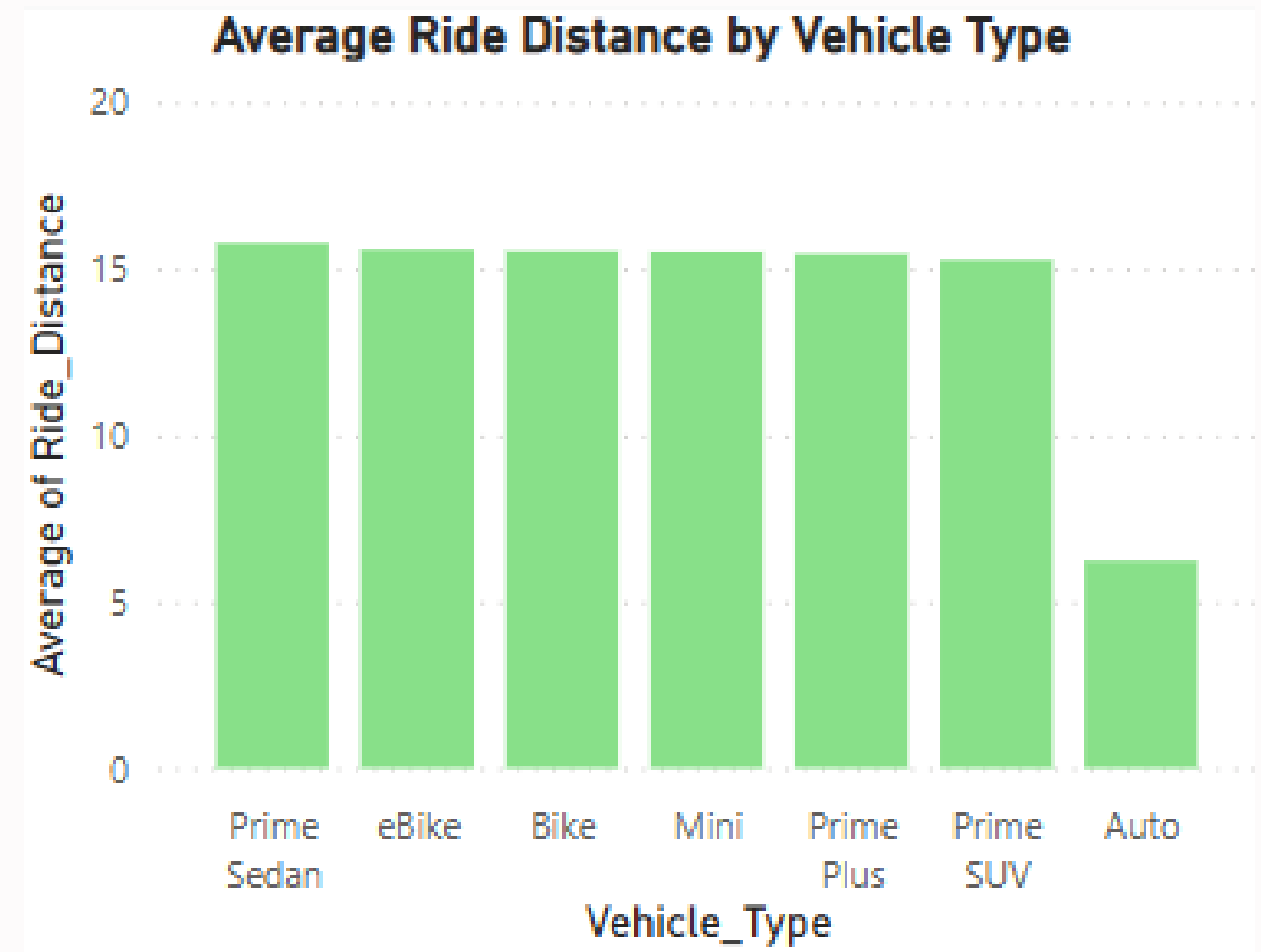
Avg Ride Distance By Vehicle Type

Insight:

- Most vehicle types have rides around 15-16 km.
- Prime Sedans cover slightly longer distances than others.
- Autos are mainly used for short trips, averaging 6.2 km.

Recommendation:

Run promotions to encourage more Auto bookings for short-distance rides, helping to capture more local trips and boost overall bookings.



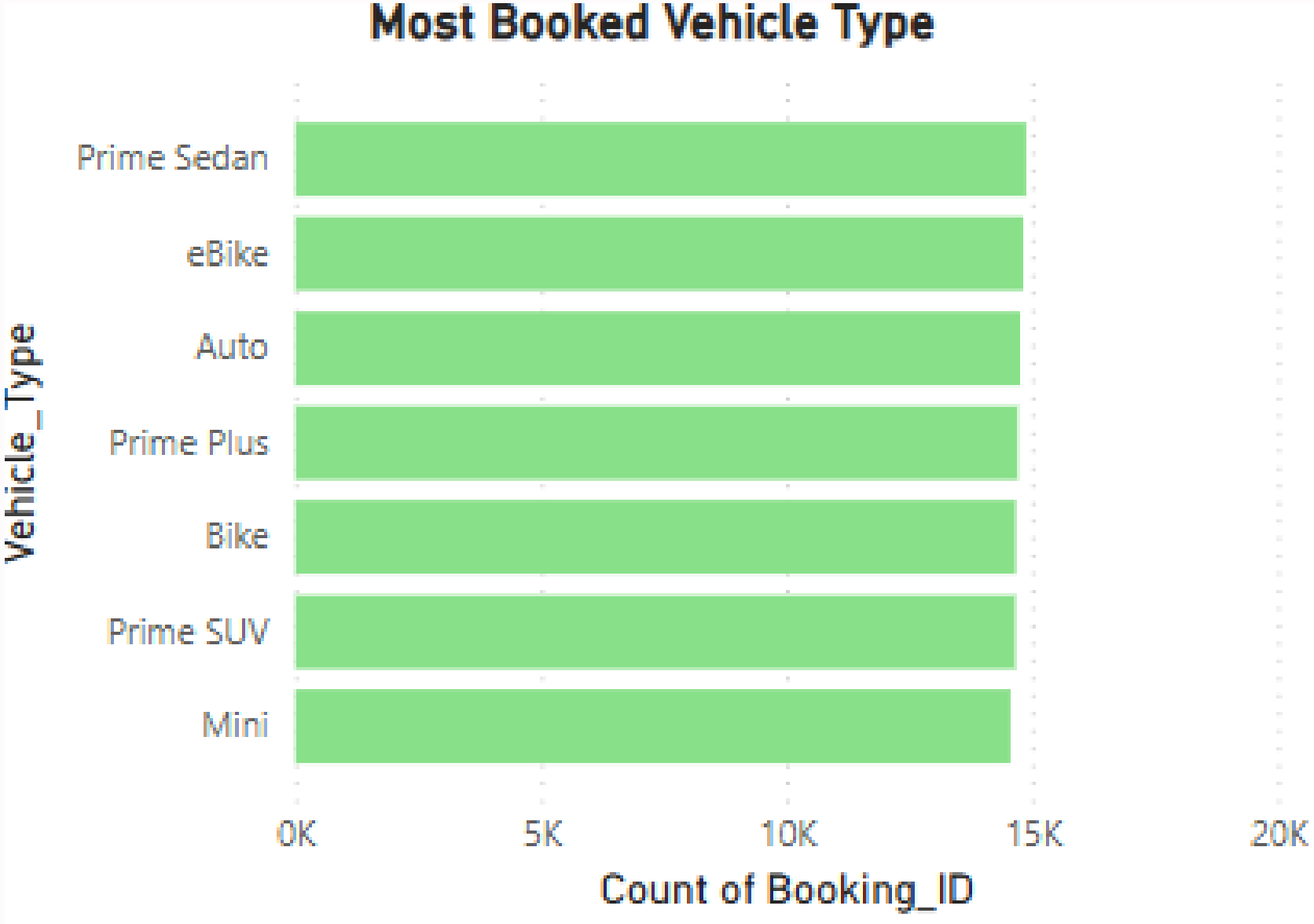
Most Booked Vehicle Type

Insight:

- Bookings are fairly similar across all vehicle types.
- Prime Sedans are booked the most, followed by E-bikes and Autos.

Recommendation:

Promote under-booked vehicle types with special offers to balance demand across categories and ensure better usage of available vehicles.



REVENUE ANALYSIS

Analyzed revenue contribution by pickup locations, payment methods, vehicle types, and identified top customers driving the highest revenue.



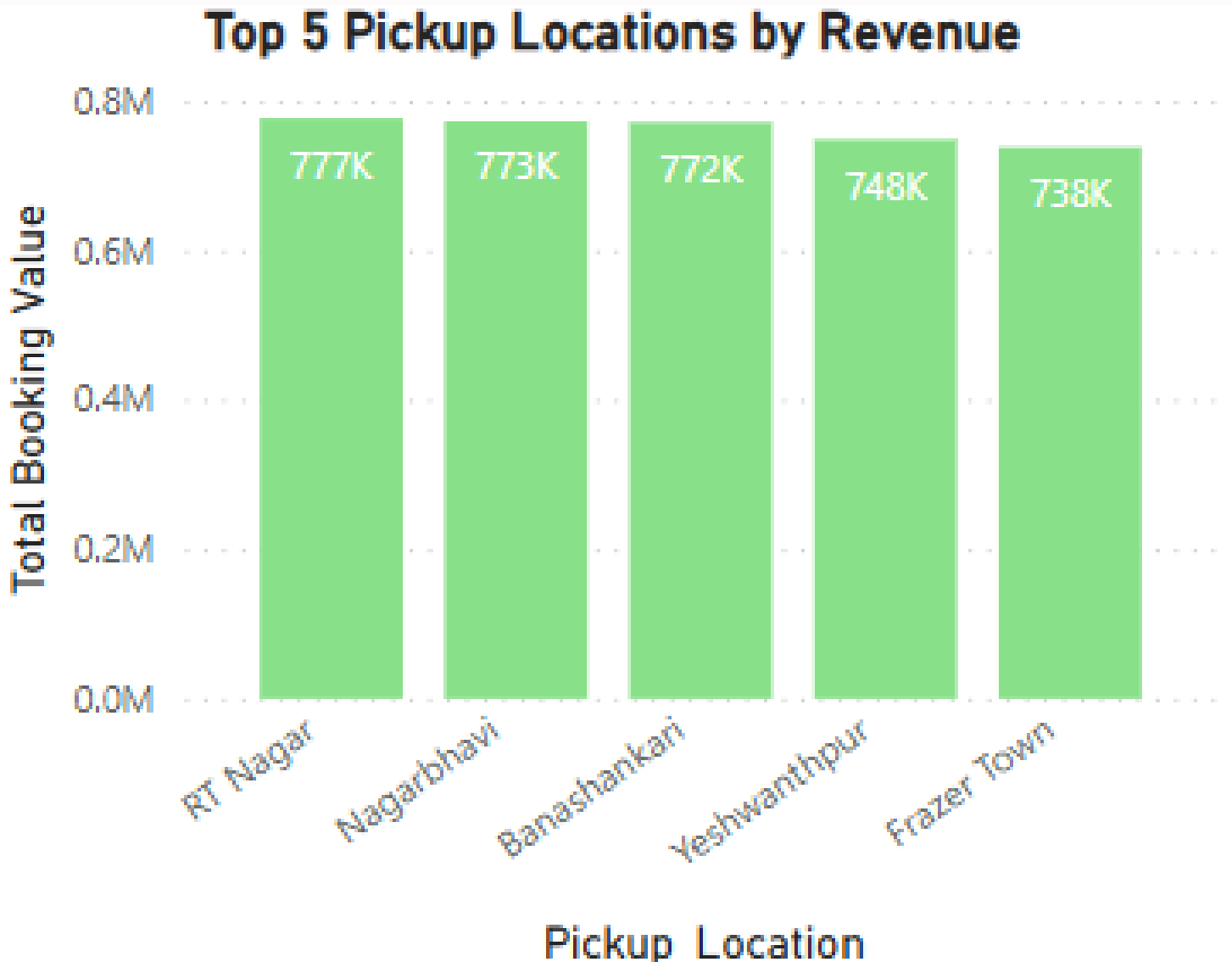
Top Pickup Locations By Revenue

Insight:

- RT Nagar, Nagarbhavi , and Banashankari generate the highest revenue.
- Yeshwanthpur and Frazer Town also perform well.

Recommendation:

Focus driver availability and promotional offers in these high-revenue zones to maximize Revenue.



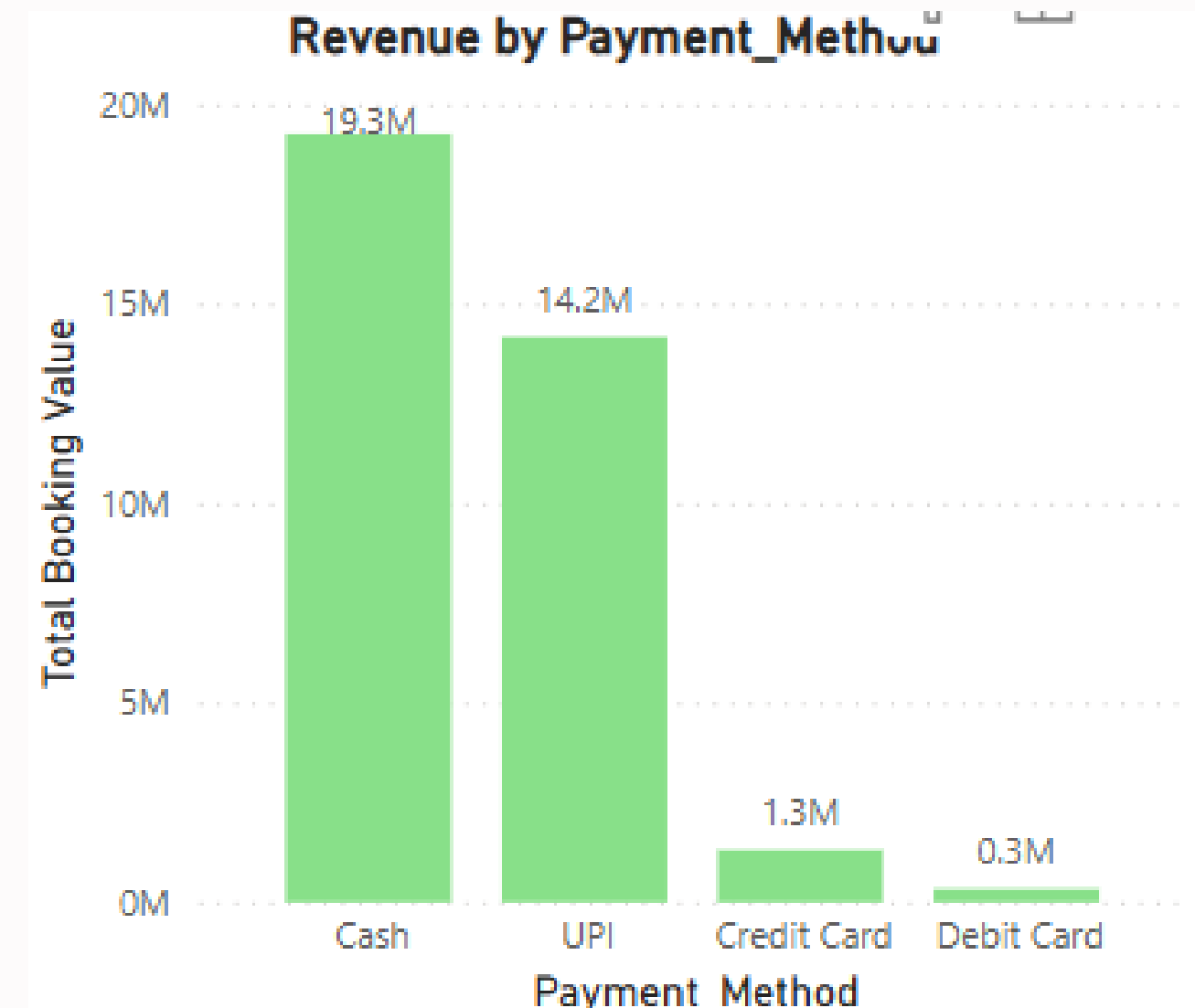
Revenue By Payment Method

Insight:

- Cash brings in most revenue (₹19 million), followed by UPI.
- Credit/Debit card usage is very low.

Recommendation:

Promote UPI and digital payments through offers or discounts to reduce cash handling and improve efficiency.



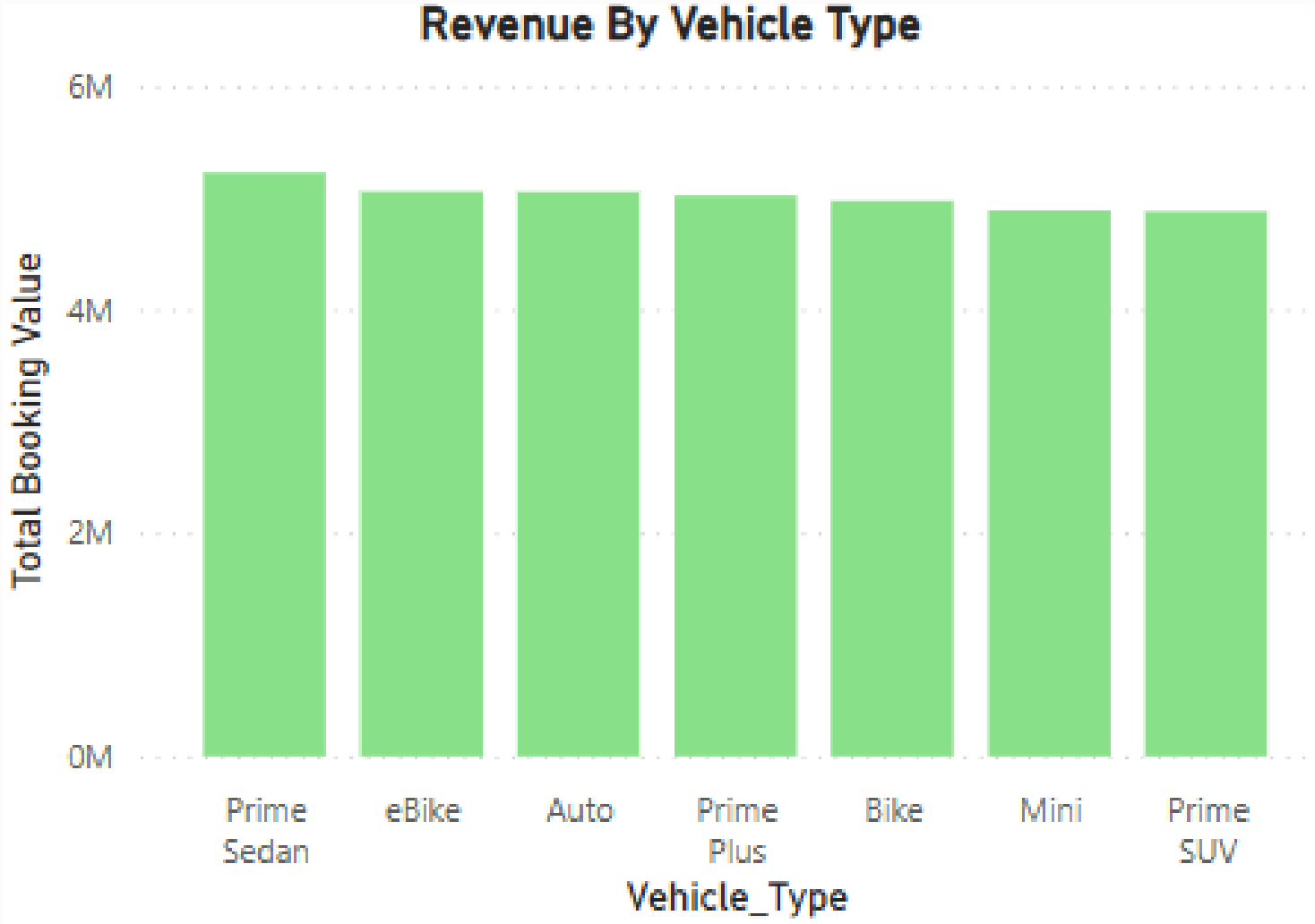
Revenue by Vehicle Type

Insight:

- All vehicle types generate similar revenue, with Prime Sedans leading slightly.
- Prime SUVs bring in the least.

Recommendation:

Keep pushing Prime Sedans with deals to maintain revenue, and review pricing or marketing for SUVs to boost demand.



Top 5 Customers by Revenue

Insight:

- CID785112, CID308763, CID734557, CID353074, CID836942
- These are the top revenue-generating customers.

Recommendation:

Focus on retaining these high-value customers and offer loyalty rewards or personalized promotions.

Top 5 Customers	
Customer_ID	Sum of Booking_Value
CID785112	8025
CID308763	6281
CID734557	6177
CID353074	6110
CID836942	6019

CANCELLATION ANALYSIS

Analyzed cancellation patterns by customers and drivers, identifying key reasons behind ride cancellations and vehicle type-related trends.



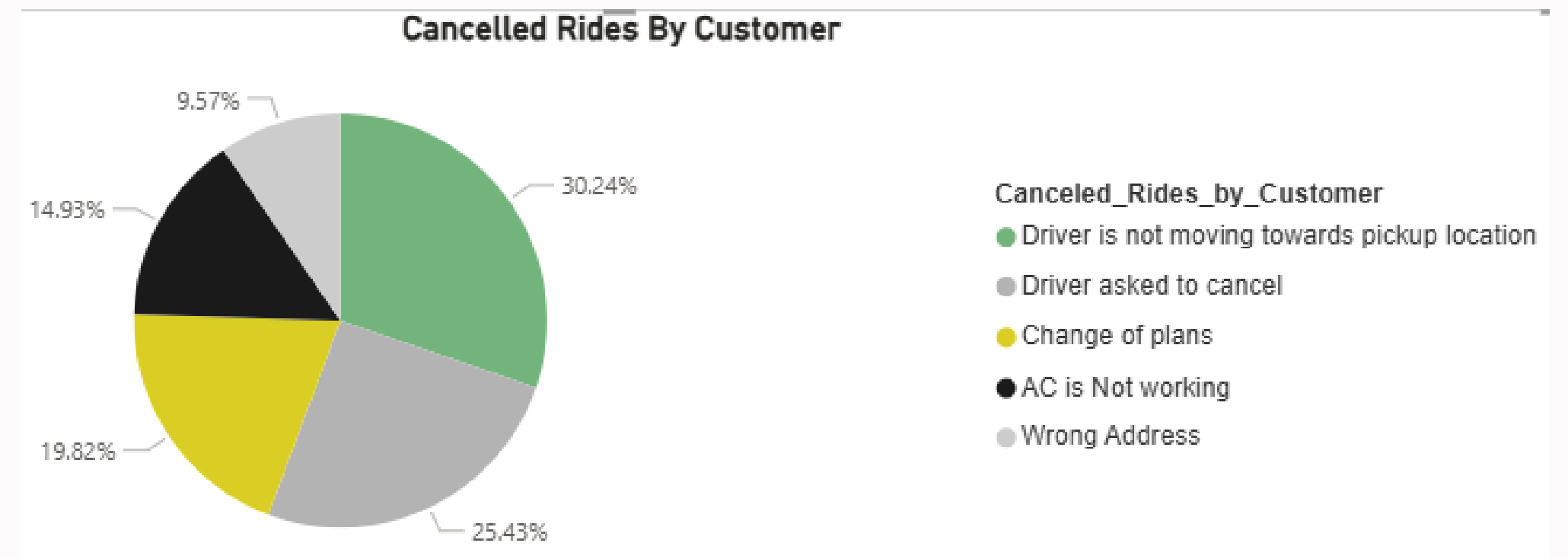
Why Customers Cancel

Insight:

- **Main reason:** 30% say drivers are not moving towards pickup.
- **Other reasons:** driver asked to cancel, change of plans, wrong address (minor cases).

Recommendation:

- Assign nearby drivers faster and ensure they start moving quickly.
- Set rules or penalties for drivers who often delay or refuse pickups



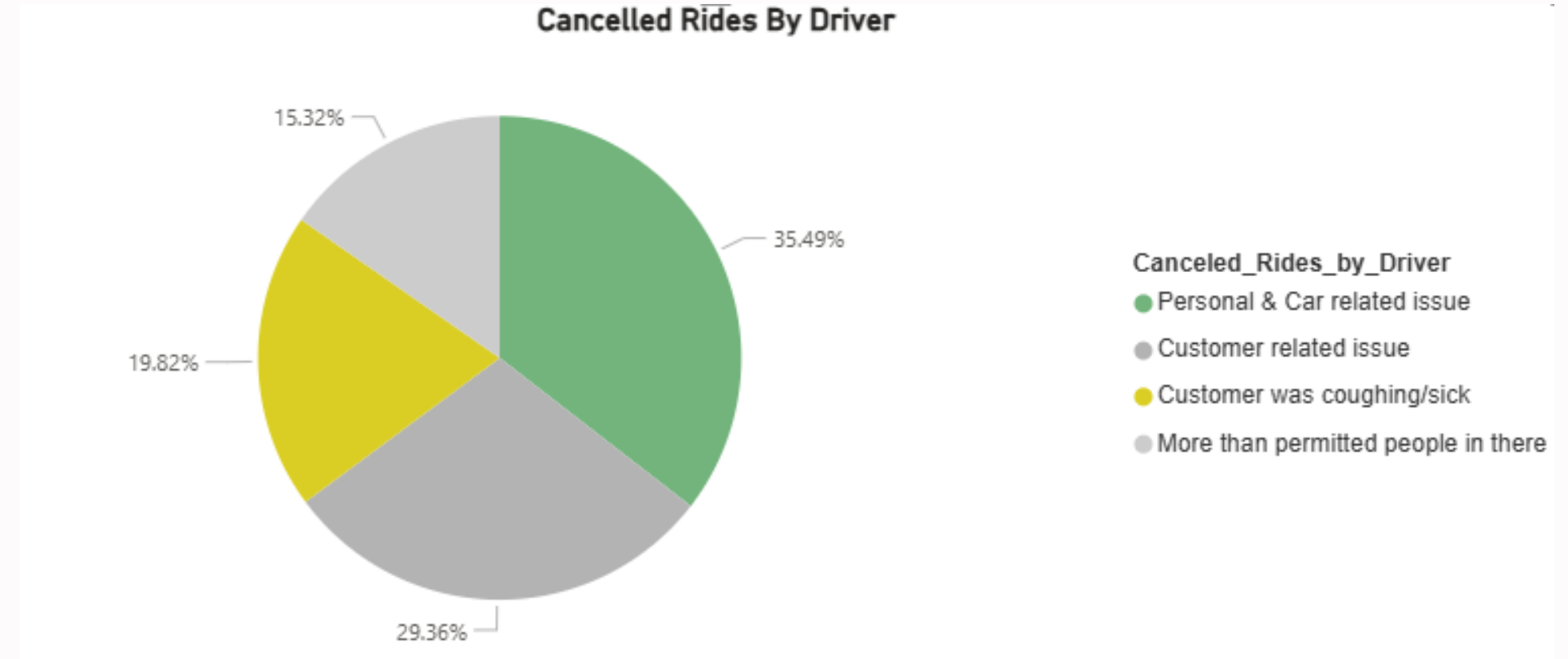
Why Drivers Cancel

Insight:

- 35% cancel because of personal or car issues.
- Other reasons: customer issues, extra passengers (rare).

Recommendation:

- Give drivers small incentives or bonuses when they complete rides without cancellations.



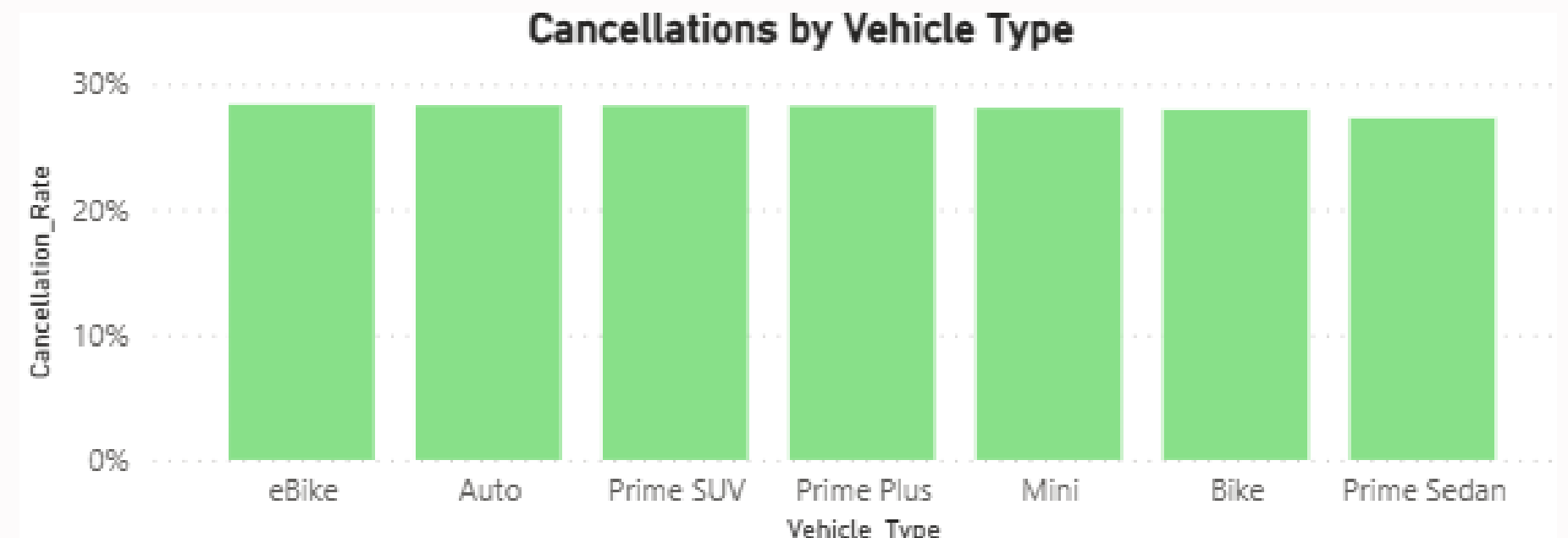
Cancellation Rate by Vehicle Type

Insight:

- Cancellation rates are similar (27-28%) across vehicle types.
- E-bikes and Autos have slightly fewer cancellations. Prime Sedans and Bikes are a bit higher.

Recommendation:

- Promote E-bikes and Autos more, especially for short trips where cancellations are fewer.
- Review and improve driver quality checks for high-cancellation vehicle types like Prime Sedans.



RATING ANALYSIS

Analyzed average driver and customer ratings across vehicle types to identify performance trends and areas for improvement.



Customer Rating Insights

Insight:

- Customers using Prime Plus give the highest ratings (4.01).
- Bike and E-bike customers give slightly lower ratings (3.99).
- Premium vehicle(prime plus) users are a little more satisfied, while bike rides show slightly lower customer experience.

Recommendation:

- Look into customer feedback on Bike rides to find small improvements and boost their ratings.

Avg Customer Ratings						
 Prime Sedan	 Prime SUV	 Prime Plus	 Mini	 Auto	 Bike	 E-Bike
4.00	4.00	4.01	4.00	4.00	3.99	3.99

Driver Rating Insights

Insight:

- All vehicle types have similar driver ratings, around 4 out of 5.
- Prime SUV and E-bikes have slightly better ratings (4.01).
- Bikes, Mini, and Sedan are a bit lower (3.99).
- Overall driver quality is consistent, but Prime SUV and E-bike drivers are doing a bit better.

Recommendation:

- Promote Prime Plus and SUV options as premium experiences with higher satisfaction.

Avg Driver Ratings						
 Prime Sedan	 Prime SUV	 Prime Plus	 Mini	 Auto	 Bike	 E-Bike
3.99	4.01	4.00	3.99	4.00	3.99	4.01



Key Insights

- ✓ 62% bookings are successful; 28% cancellations — mostly due to driver issues.
 - ✓ Customer demand is the top reason for incomplete rides; vehicle breakdowns come next.
 - ✓ CID785112 has the highest cancellations and is also a top revenue customer.
 - ✓ Weekdays see higher ride demand; weekends are lower.
 - ✓ Banashankari, Yeshwanthpur, RT Nagar are top pickup zones.
 - ✓ Peenya, Mysore Road, MG Road are top drop zones.
 - ✓ Prime Sedans & E-bikes are the most booked vehicles.
 - ✓ Prime Sedans & Prime Plus generate the highest revenue per ride.
 - ✓ RT Nagar, Nagarbhavi, Banashankari are top revenue-generating zones.
 - ✓ Cash is the main payment mode; UPI is growing.
 - ✓ Driver ratings are consistent (~4★); Prime SUV & E-bike drivers rated slightly higher.
 - ✓ Prime Plus users give the highest customer ratings.
- 



Business Recommendations

- ✓ Reduce cancellations: Incentivize drivers, apply penalties for frequent cancellations.
- ✓ Simplify rescheduling options for customers to lower demand-related cancellations.
- ✓ Promote Prime Sedans & Prime Plus with offers to maximize high-revenue bookings.
- ✓ Boost Autos & E-bikes for short trips with fewer cancellations.
- ✓ Focus driver availability & offers in RT Nagar, Nagarbhavi, Banashankari to capture revenue.
- ✓ Push UPI/digital payments with small discounts to reduce cash handling.
- ✓ Monitor and retain top customers (like CID785112) with loyalty offers, while controlling their cancellations.
- ✓ Analyze Bike ride feedback to improve customer ratings and satisfaction.

THANK YOU

